



Radiologists' perceptions on AI integration: An in-depth survey study

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ABSTRACT

Purpose: To assess the perceptions and attitudes of radiologists toward the adoption of artificial intelligence (AI) in clinical practice.

Methods: A survey was conducted among members of the SIRM Lombardy. Radiologists' attitudes were assessed comprehensively, covering satisfaction with AI-based tools, propensity for innovation, and optimism for the future. The questionnaire consisted of two sections: the first gathered demographic and professional information using categorical responses, while the second evaluated radiologists' attitudes toward AI through Likert-type responses ranging from 1 to 5 (with 1 representing extremely negative attitudes, 3 indicating a neutral stance, and 5 reflecting extremely positive attitudes). Questionnaire refinement involved an iterative process with expert panels and a pilot phase to enhance consistency and eliminate redundancy. Exploratory data analysis employed descriptive statistics and visual assessment of Likert plots, supported by non-parametric tests for subgroup comparisons for a thorough analysis of specific emerging patterns.

Results: The survey yielded 232 valid responses. The findings reveal a generally optimistic outlook on AI adoption, especially among young radiologist (<30) and seasoned professionals (>60, $p < 0.01$). However, while 36.2 % (84 out of 232) of subjects reported daily use of AI-based tools, only a third considered their contribution decisive (30 %, 25 out of 84). AI literacy varied, with a notable proportion feeling inadequately informed (36 %, 84 out of 232), particularly among younger radiologists (46 %, $p < 0.01$). Positive attitudes towards the potential of AI to improve detection, characterization of anomalies and reduce workload (positive answers > 80 %) and were consistent across subgroups. Radiologists' opinions were more skeptical about the role of AI in enhancing decision-making processes, including the choice of further investigation, and in personalized medicine in general. Overall, respondents recognized AI's significant impact on the radiology profession, viewing it as an opportunity (61 %, 141 out of 232) rather than a threat (18 %, 42 out of 232), with a majority expressing belief in AI's relevance to future radiologists' career choices (60 %, 139 out of 232). However, there were some concerns, particularly among breast radiologists (20 of 232 responders), regarding the potential impact of AI on the profession. Eighty-four percent of the respondents consider the final assessment by the radiologist still to be essential.

Conclusion: Our results indicate an overall positive attitude towards the adoption of AI in radiology, though this is moderated by concerns regarding training and practical efficacy. Addressing AI literacy gaps, especially among younger radiologists, is essential. Furthermore, proactively adapting to technological advancements is crucial to

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fully leverage AI's potential benefits. Despite the generally positive outlook among radiologists, there remains significant work to be done to enhance the integration and widespread use of AI tools in clinical practice.

1. Introduction

Thanks to the increasing knowledge, data availability, and computation power, medical imaging stands at the frontier of innovation for the development and application of artificial intelligence (AI)-based tools, potentially revolutionizing the radiologist's role [1,2,3,4]. However, the actual implementation of AI into clinical practice has yet to keep pace with scientific advancements [5,6], leaving a lingering debate regarding its risks and benefits [7].

The successful integration of AI tools into clinical workflows hinges upon the active engagement of healthcare professionals, who serve as primary stakeholders interfacing with AI systems daily [8]. Consequently, understanding the perceptions and attitudes of radiologists towards this transformative technology emerges as a critical imperative.

Recent studies probing the opinions of medical imaging personnel on AI in medicine have revealed a generally positive outlook [2,9,10]. Nevertheless, concerns surrounding technical expertise, ethical considerations, and professional responsibilities persist. Surveys, such as the one conducted by the American Society of Radiologic Technologists in 2019 [11], have underscored apprehensions about potential impacts on the "human" aspects of the profession, including patient interaction. Similarly, insights gleaned from a survey of members of the European Society of Radiology have highlighted both the potentially beneficial impacts of AI in making radiology more clinically and patient-oriented, alongside contentious perspectives on liability issues [2].

Some authors have identified various barriers to AI implementation in medicine, ranging from concerns about AI-related errors and cybersecurity to data protection and knowledge gaps [12], suggesting that the slow adoption of AI systems in medical practice may be exacerbated by insufficient education and unclear regulations. Therefore, physicians must be aware of the challenges and limitations of artificial intelligence to lead its effective application into clinical routines [5,12].

This survey-based investigation aims to explore the radiologists' opinions on the application of AI-based tools in clinical practice. By centering on the attitudes of healthcare professionals, this inquiry seeks to enlighten pathways for overcoming barriers to the widespread adoption of AI technologies within radiology.

2. Methods

2.1. Questionnaire design and item selection

The questionnaire was designed to explore participants' opinions and attitudes regarding current and future applications of AI in radiology, along with their relationship to certain "latent" variables, including demographic and professional characteristics. Radiologists' attitudes were considered broadly and from different perspectives, including satisfaction with the use of AI-based tools, propensity for innovation, and optimism for the future.

The questionnaire was divided into two sections. The first part collected demographic and professional information and required categorical responses. The second part aimed at assessing respondents' opinions on AI and included Likert-type responses with values ranging from 1 to 5 (except question Q2).

Since ours is an opinion survey and the entity measured is not a strictly psychological construct, the standard approach used for the Likert questionnaire creation was adapted to our case [13].

All questions were formulated so that the value "3" indicated a neutral attitude, higher values on the scale (4–5) represented increasingly positive attitudes (i.e. greater trust or optimism towards AI), and vice versa for values 2–1. The only exception was question Q12 ("How

important do you consider the final assessment of the AI results by the radiologist?") because it was believed that reversing the question might confuse the reader.

The preliminary phase of the questionnaire's formulation followed a MiniDelphi-inspired approach for the selection of relevant topics [15,15,16], involving a panel of experts previously consulted for another study [17], including some of the authors (S.I., A.F., C.F., M.C., M.A.), radiologists of the Regional Committee of the Regional Group of Lombardy of the Italian Society of Medical and Interventional Radiology (SIRM Lombardy, Milan, Italy), and experts in Human-Machine Interaction of the Department of Educational, Psychological and Communication Sciences and of the Department of Legal Sciences of the University of Suor Orsola Benincasa (Naples, Italy). In addition, we also considered a similar survey published by the European Society of Radiology [18].

An initial pool of 50 questions including demographic and professional data was collected and, after a preliminary screening carried out by the panel of experts by consensus, a first version of the questionnaire was tested on a pilot sample ($n = 25$) of radiologists and residents in radiology to eliminate ambiguity, verify understanding, and evaluate time to completion.

The further selection process was undertaken to halve the questionnaire's size and reduce completion time to < 5 min to ensure recipient compliance.

To identify potentially redundant questions, we analyzed the responses through a correlation matrix, and adopting a stepwise backward selection for each pair of questions we discarded the item with a lower variance. If the items were highly interrelated but the authors believed they provided independent information, both questions were retained.

In formulating the final questionnaire, we also considered whether the items were arranged in a logical sequence and whether there was expectedness between responses, aiming to ensure the overall coherence of the questionnaire.

To ease the administration and the analysis the questionnaire included only closed, concise questions expressed in plain language, avoiding double-negative formulations.

2.2. Sample size estimation

The questionnaire was initially planned as an opinion survey designed to describe the existing distribution of variables, without regard to causal or other specific hypotheses defined a priori. Therefore, to estimate the sample size we first started considering the target population and followed some general well-known approaches like items-sample ratios and other rules-of-thumb [19].

In addition, in the event of conducting a posteriori comparison between subgroups of responders motivated by the exploratory data analysis, an *a priori* power analysis was also conducted adopting the open-source software G*power (Version 3.1.9.6) [20] using the following parameters: effect size 0.15 (medium), $\alpha = 0.05$, 20 predictors, and power = 0.80 [21], yielding a sample size estimation of 157 subjects.

In conclusion, a desired sample size equal to 10 % of our reference population (around 2000 subjects) was judged suitable for our purpose and a conservative target size of 220 subjects was finally established.

2.3. Survey administration and data collection

Our target population consists of radiologists operating in the region of Lombardy (Italy) registered in the Italian Society of Medical and Interventional Radiology (SIRM Lombardy, Milan, Italy). Each member

received a personal e-mail invitation to respond to the survey, which could be accessed only once over approximately 2 months, from 12/10/2023 to 22/01/2024. Two reminders were sent one week before and on the final day.

The questionnaire was available in Italian language using Survey Monkey (<https://www.monkeysurvey.com>, Momentive Inc., San Mateo, California, USA) and structured as clearly and lightly as possible to allow responding in a few minutes and to be able to get the opinions of as many radiology physicians as possible.

The survey package included a short cover letter to present the questionnaire.

2.4. Final version of the questionnaire

The final version of the questionnaire (**Supplementary Material 1**) comprised a total of 21 items divided into two sections: general questions (G1-G7) and AI-related questions (Q1-Q14).

The first section focused on the demographic and professional characteristics of the respondents including gender, age, role (specialist, resident), seniority, type of contract, main radiological activity (sub-specialty), and work context (public hospital, private centers, etc.). The continuous variables “age” and “seniority” were discretized into 4 and 3 classes respectively.

The second section comprised 14 questions aimed at evaluating radiologists’ attitudes towards the application of AI in radiology. Initially, radiologists were asked whether they currently use AI tools in their daily practice (Q1). The next two questions were conditional (Q2, Q3): only those who had answered affirmatively to question Q1 responded to question Q2 (“Do you think they could be useful in your daily clinical practice?”), while only those who had answered “No” or “I don’t know” responded to question Q3 (“Do you think that AI-based systems are of effective support to you in your daily clinical practice?”).

Subsequently, the radiologists were asked to self-assess their level of awareness regarding AI in general (Q4).

The responses to these previous questions have been also evaluated as independent predictors for the other questions.

Questions Q5-Q7 aimed to assess the role of AI in supporting interpretative and diagnostic aspects, particularly regarding detection (Q5), characterization (Q6), and the choice of further investigation (Q7).

The subsequent group of questions, Q8-Q10, focused on evaluating the role of AI from the perspective of personalized medicine through the reduction of radiation exposure (Q8), the choice of investigation protocol (Q9), and “personalization” of the radiological investigation in general (Q10).

The last four questions concerned the relationship between AI-based tools and the profession of radiologist, specifically: the role of artificial intelligence in improving workflow and reducing workload (Q11), the role of the radiologist as the final supervisor of the results of AI (Q12), the importance of AI shortly on the attractiveness of specializing in radiology and, finally, the judgment on the degree of threat represented by AI for the radiology profession (Q14).

2.5. Exploratory data analysis

Survey responses were automatically recorded in Excel (Microsoft, Redmond, WA, USA). Data cleaning, graphing, and statistical analysis were performed using R (version 2024.04.1 + 748; R Core Team, 2024).

Standard descriptive statistics were calculated for demographic variables and Likert-like responses. Questions from Q3 to Q14 were used to assess respondents’ attitudes towards adopting AI-driven tools in radiology. To visualize the overall trend of the pooled population, the average responses of each participant, rounded to the nearest unit, were displayed in a Likert plot. Question 12 (“How important do you consider the final evaluation of the AI results by the radiologist?”) was excluded due to its potential ambiguity regarding AI adoption.

Demographic and professional variables were visualized using

histograms. Five-point Likert scale data were depicted with divergent stacked bar graphs to illustrate central tendencies for enhanced data exploration. Likert plots were designed both for the entire pool of respondents and stratified according to demographic and professional variables (**Supplementary Material 2**).

A visual inspection of all Likert plots was conducted; where the patterns suggested significant differences among respondent subgroups, additional statistical analyses were performed. Clues for data exploration also came from the panel of experts and the authors of the study based on the discussions developed in the preliminary phase.

The non-parametric Mann-Whitney U and Kruskal-Wallis tests were used to compare Likert ordinal responses between subgroups, respectively for bimodal and multilevel categorical variables. Subsequent pairwise post-hoc comparisons adjusted for multiple testing were conducted upon identifying significant results. Given that the “<30” age group was overrepresented, we addressed the issue of class imbalance by oversampling the minority age categories and retested the significant findings when stratifying by age. (See **Fig. 1**).

3. Results

3.1. Study population

240 radiologists (out of 2000 members of SIRM Lombardy) responded to the survey, however, 8 of them did not complete the entire questionnaire, resulting in a total of 232 responders eligible for the analysis (response rate 42 %). Demographic and professional data are presented in **Fig. 2**.

3.2. General trend on the adoption of AI in radiology (Q1-Q3)

Q1 – Do you currently use AI-based tools in your clinical practice?

Q2 – Do you think that AI-based software could be useful in your daily clinical practice?

Q3 – Do you think that AI-based systems are of effective support to you in your daily clinical practice?

Eighty-four subjects (36.2 %) reported using AI-based tools in clinical practice daily, 38.8 % stated they do not use them, and 36.2 % were unsure if they use them (Q1) (**Fig. 3**).

Among respondents currently using AI tools in clinical practice (Q2), opinions are nearly evenly divided: 27 % believe these tools do not make an effective contribution, 30 % consider their contribution decisive, and the remaining 43 % express a neutral opinion (**Fig. 4**). The majority of daily users work with CT (28.5 %) and MRI (22.5 %). Conversely, 85 % of those who do not currently use AI-driven tools (Q3) believe they would benefit from these tools if available in their clinical practice (**Fig. 4**).

3.3. Overall attitude towards AI

Questions from Q3 to Q14 (excluding Q12) were averaged to represent respondents’ overall attitudes towards AI-based tools. **Fig. 5** illustrates the distribution of these mean Likert responses, revealing that 68 % of respondents demonstrated a generally optimistic attitude. When stratified by age, both younger (<30 years) and older (>60 years) radiologists demonstrated more positive attitudes towards AI adoption, with similar means and high percentages of positive opinions. Young radiologists are those who present the highest percentage of positive opinions (**Fig. 5**, bottom row, 76 %). The intermediate age groups (30–45 and 45–60 years) exhibited less enthusiastic attitudes towards AI adoption. Notably, the 45–60 age group demonstrated the greatest balance between positive and negative judgments, along with the highest variability in responses. This group’s average responses were significantly lower compared to other age groups, as confirmed by the post-hoc Dunn’s test adjusted for multiple comparisons ($p < 0.01$).

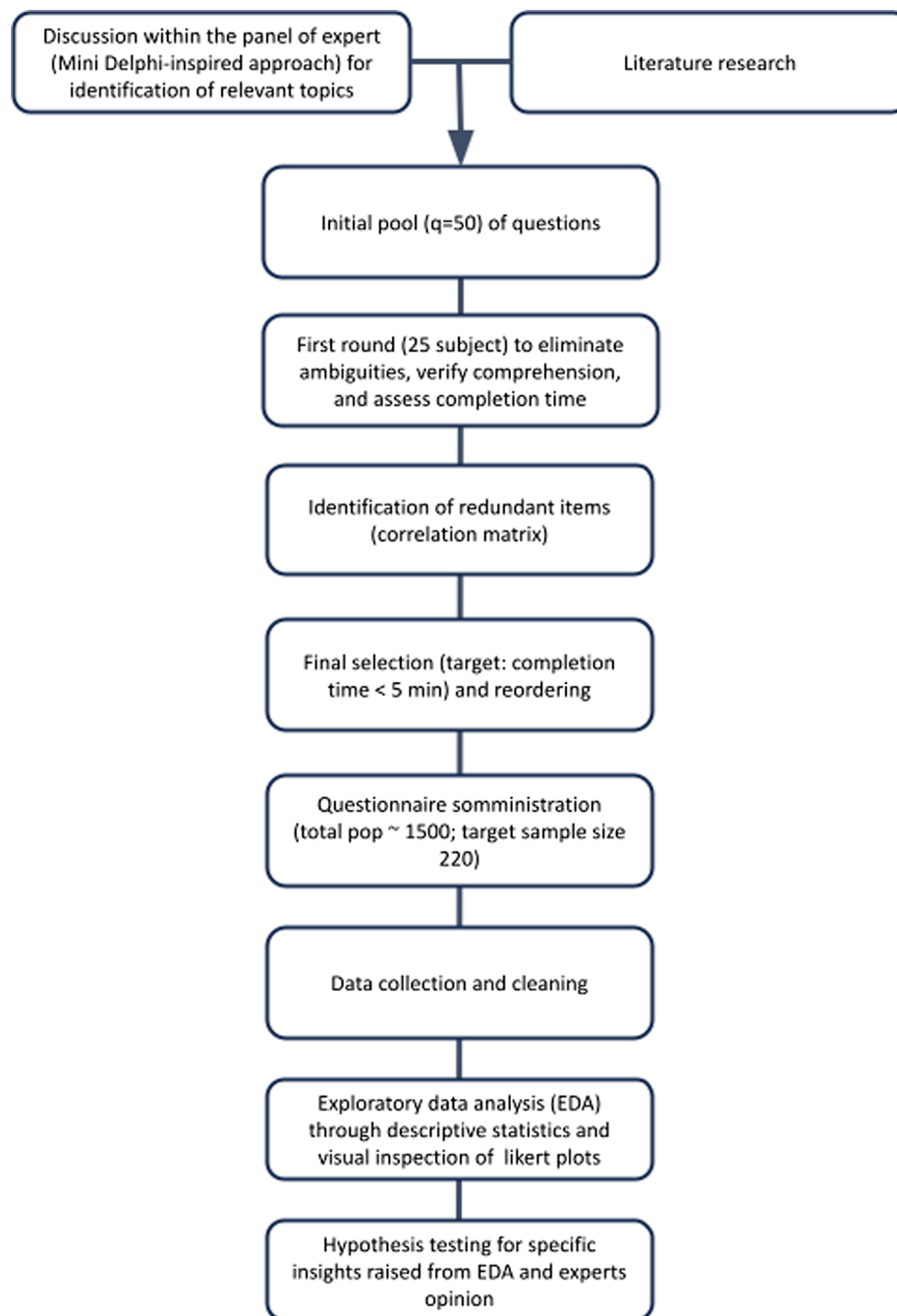


Fig. 1. Flowchart describing the questionnaire design and item selection.

3.4. Subgroup analysis

Upon closer examination of the Likert plot in Fig. 6, the question regarding the self-assessment of AI literacy (Q4) revealed predominantly negative opinions, with 78 % of respondents rating their knowledge on the topic as inadequate to neutral, and 36 % specifically expressing “none” or “very poor” knowledge.

The distribution of responses to other questions reflects a generally favorable attitude towards the potential of AI and its implementation in clinical practice. For instance, over 80 % of interviewees expressed favorable opinions on AI’s role in improving lesion detection (Q5), characterization (Q6), and reducing the radiologist’s workload (Q11). However, more conflicting opinions were observed regarding AI’s potential in assisting with the choice of in-depth investigations (Q7), improving protocol customization (Q9), and offering a personalized

service based on patient needs (Q10). As regards the importance of the final evaluation by the radiologist, it remains fundamental for 84 % of interviewees (Q12).

Q4 – “How informed do you consider yourself on the topic of AI in the healthcare and specifically in radiology?”.

Residents rate their knowledge of AI significantly lower than established specialists ($p < 0.01$). Thirty-four percent of specialists consider themselves “very” or “fairly” informed compared to only 10 % of residents, while only 26 % of specialists consider themselves “not very” or “not at all” informed compared to 46 % of trainees (Fig. 7). Comparing responses to question Q4 by age groups reveals significant differences ($p < 0.01$), especially between those under 30 (mostly residents) and those over 60.

Stratifying for prevalent activity shows that in general those who mainly carry out traditional radiology considered themselves less informed

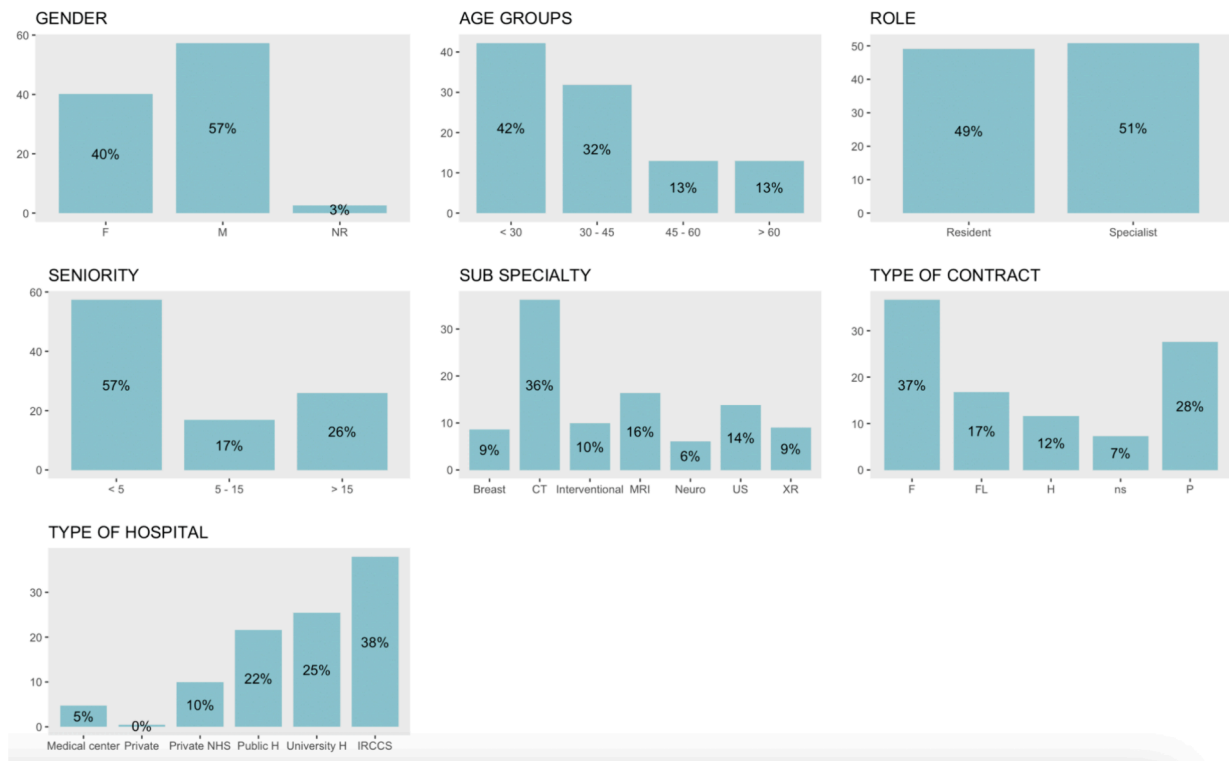


Fig. 2. Demographic and professional characteristics of the sample.

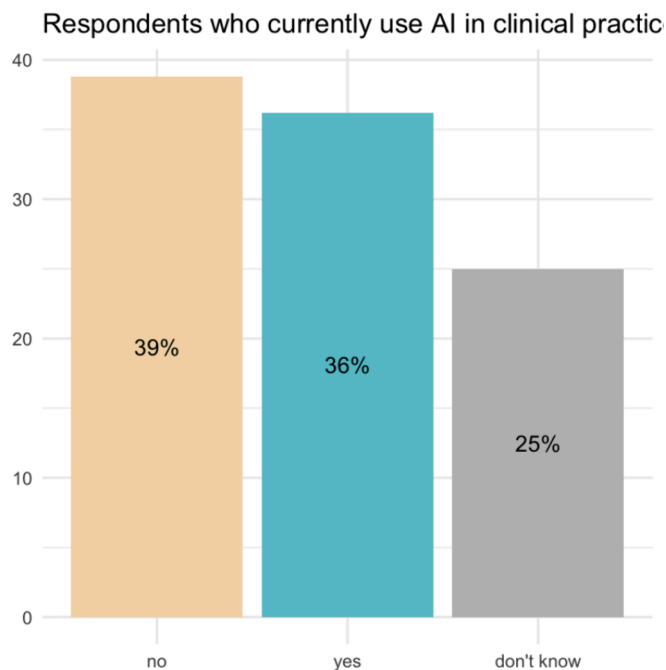


Fig. 3. Distribution of responses to question Q1.

(52 %) while the total of opinions 4 and 5 is highest among neuroradiologists (50 %).

Q5 – “Can AI-driven systems assist radiologists in the detection of abnormalities in diagnostic examinations?”.

Q6 – “Can AI-driven systems improve diagnostic accuracy (in extremis, lesion characterization)?”.

As regards the role of AI in the detection (Q5) and characterization (Q6) of abnormalities, the opinion was generally positive among all

subgroups of respondents, regardless of age group and subspecialty (Fig. 8), with radiologists dealing with traditional radiology being slightly more pessimistic regarding the detection task (sum of negative opinions 19 %). Overall the radiologists expressed favorable opinions on both questions.

Q7 – “Can AI-driven systems improve the radiologist’s decision-making processes (for example the choice of the most appropriate imaging method/interventional procedure)?”.

Only 50 % of responders believe that AI-based systems can improve the radiologist’s decision-making processes and in particular the choice of subsequent investigations (Q7).

Q8 – “Can AI-driven systems optimize radiation exposure for increasingly safer diagnoses?”.

Q9 – “Can AI-driven systems assist the radiologist in selecting and adapting the investigation protocol based on patient characteristics?”.

Q10 – “Can AI-based systems allow the ‘personalization’ of patient management and diagnosis?”.

Questions Q8-Q10 were centered on the role of AI in optimizing the radiological investigation concerning the patient’s characteristics. Homogeneous opinions are instead obtained in response to question Q8 regarding dose reduction, with 68 % of the total interviewees expressing a positive opinion on the potential role of AI in reducing radiation exposure, with interventional radiologists being more optimistic (87 %), while breast specialists are the least optimistic (25 % negative opinions) (Fig. 9).

More contrasting opinions are obtained concerning questions Q9 and Q10, respectively on the role of AI in protocol optimization and the personalization of the radiological investigation in general. Regarding the choice of protocol, the highest percentage of favorable opinions is recorded among those who mainly deal with traditional radiology 81 %, while those who deal with MR are more skeptical, with a clear prevalence for a neutral attitude (39 %).

Q11 – “Can AI-driven systems optimize the workload and facilitate the radiologist’s daily activity?”.

Q12 – “How important is the final evaluation of the AI results by the

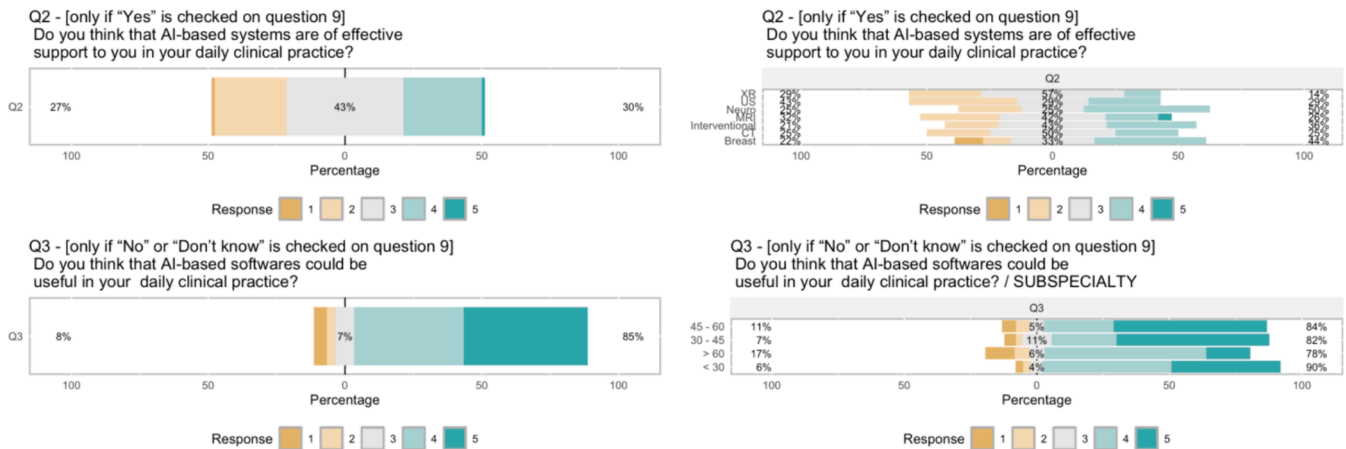


Fig. 4. Responses to question Q2 (upper row) and Q3 (bottom row) for the total sample and stratified by subspecialty.

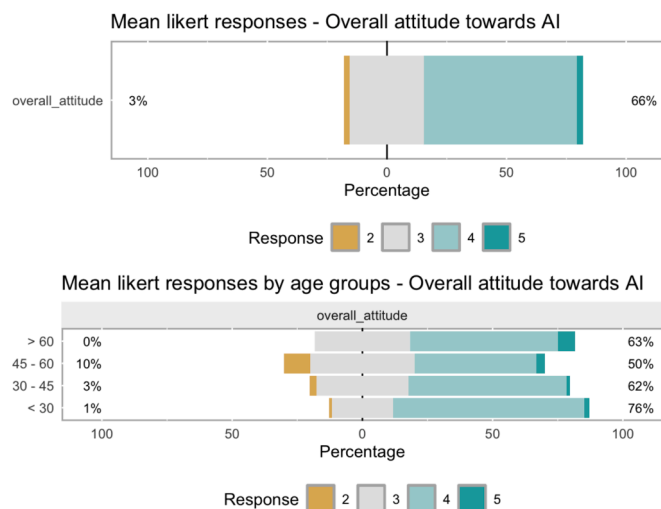


Fig. 5. First row: Likert plot illustrating the distribution of averaged responses across the total sample. Second row: Averaged responses stratified by age group.

radiologist?",

Q13 – "Do you believe that AI is a threat (=1) or an opportunity (=5) for the radiologist profession?",

Q14 – "Do you think that in the future the choice of specialization in radiology will be influenced by the growing application of AI?".

Questions (Q11-Q14) concerned the relationship between AI and the profession of radiologist.

As regards the possibility that AI assists the radiologist in improving workflow (Q11), there is generally a positive attitude across all age groups, regardless of the type of activity (Fig. 10).

The final evaluation of the AI results by the radiologist is considered of fundamental importance by over 80 % of radiologists with homogeneous distributions between the different categories of interviewees (Fig. 11) and a peak of 100 % in the subcategory of breast radiologists.

The question of whether AI represents a threat or an opportunity for the radiologist is a crucial issue in today's debate and was explored in depth with question Q13. Overall, the results suggest an optimistic attitude supporting the idea that AI can represent an opportunity (61 % of scores 4–5) rather than a threat, with a relatively homogeneous distribution of responses among the different subgroups (Fig. 12). Breast radiologists are those who express greater concern about the advent of AI compared to other subspecialties, split equally between positive and negative opinions (40 %), although this result does not reach statistical

significance ($p = 0.2375$).

Finally, participants were asked how relevant they consider the role of AI in determining the choice to enroll in the radiology specialization school shortly. Also in this case the results are rather homogeneous, with 60 % of respondents believing that the topic of AI will have a fairly or very decisive weight in the choice of future radiologists, an opinion shared by all subgroups.

4. Discussion

In the medical field, radiology is at the forefront of integrating AI technologies [14]. Several AI-driven models already demonstrated promising capabilities in abnormality detection and characterization, improving clinical outcomes with an increasingly personalized approach [22]. However, clinical translatability remains slow due to several factors, including varying attitudes towards innovation, lack of specific training, organizational challenges, and medico-legal concerns. For this reason, our study aimed to investigate how radiologists perceive the advent of AI and its clinical implications. Firstly, a generally positive attitude towards AI adoption has emerged, with a notable U-shaped distribution showing the most positive scores among young radiologists (<30) and older ones (>60). A first hypothesis is that young radiologists, being digital natives, may view AI as an ally to enhance their diagnostic skills, temporarily setting aside concerns related to professional implications. On the other hand, older radiologists may see AI as a means to improve efficiency and reduce repetitive tasks. In contrast, intermediate age groups (30–45 and 45–60 years) might be more resistant to change due to established practices, perceiving AI as both a potential threat and a cautious opportunity [1,22]. This nuanced view highlights the need for tailored support and training to address the specific concerns and preferences of each age group in the context of AI adoption.

4.1. Ai-based tools and radiologists' satisfaction in clinical practice

The implementation of AI-based tools is slowly gaining traction in clinical practice. In a survey conducted by the European Society of Radiology in 2018 only 20 % of respondents were currently using AI-based tools [2], but in a new survey of the same population in 2022 the percentage of users rose to 40 % [23,24]. However, despite its great potential, the integration of these tools is slow when compared to the exponential growth of literature on the topic [25]. The extent of this phenomenon is well illustrated by the results of our survey, which show that only 36 % of responders currently use AI in their clinical practice, results in line with those published by Tajaldeen et al. [26] and Waymel et al. [27]. As observed by some authors, the slowdown in the adoption of these tools is not so much linked to technological limitations, but also to issues such as costs, medico-legal and ethical concerns [1,28], and the

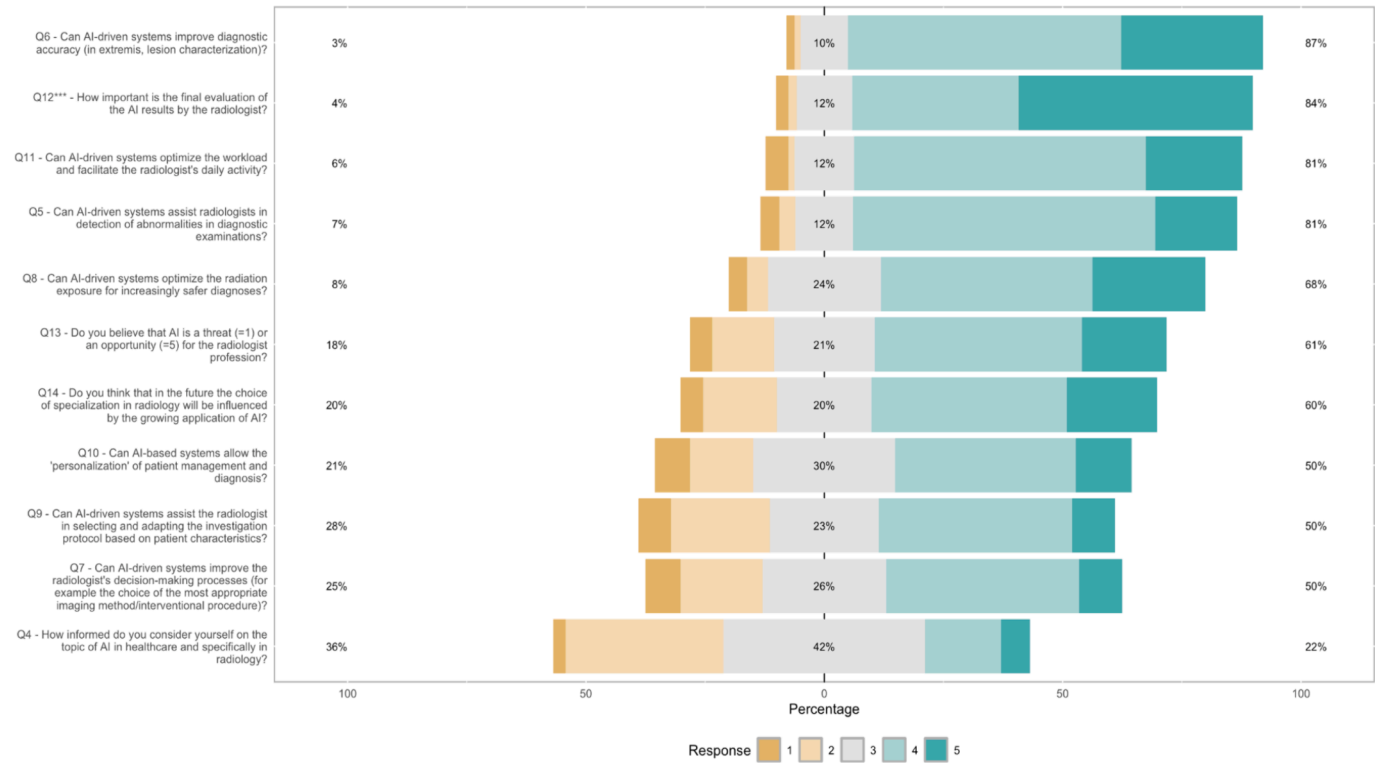


Fig. 6. Likert plot representing the distribution of responses to each question Q3-Q14.

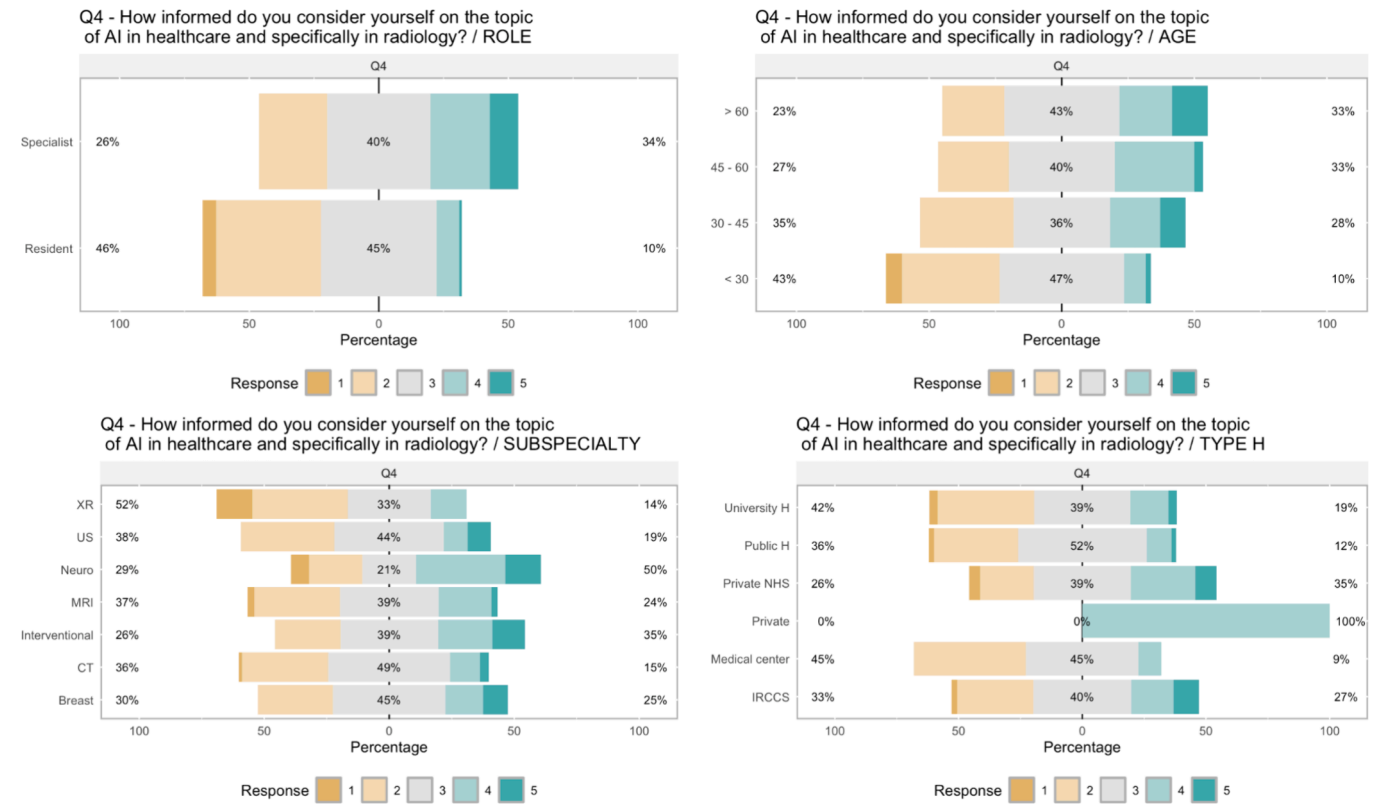


Fig. 7. Responses to question Q4 stratified by role, age, subspecialty, and type of hospital.

training of operators. Noteworthy is the difference in opinion between those who regularly use such tools and those who do not. Interestingly, the vast majority of

those who do not currently use AI in their clinical routine believe it could be highly effective, if not essential, in daily practice. Conversely, among those who currently use AI, only 30 % of respondents believe that

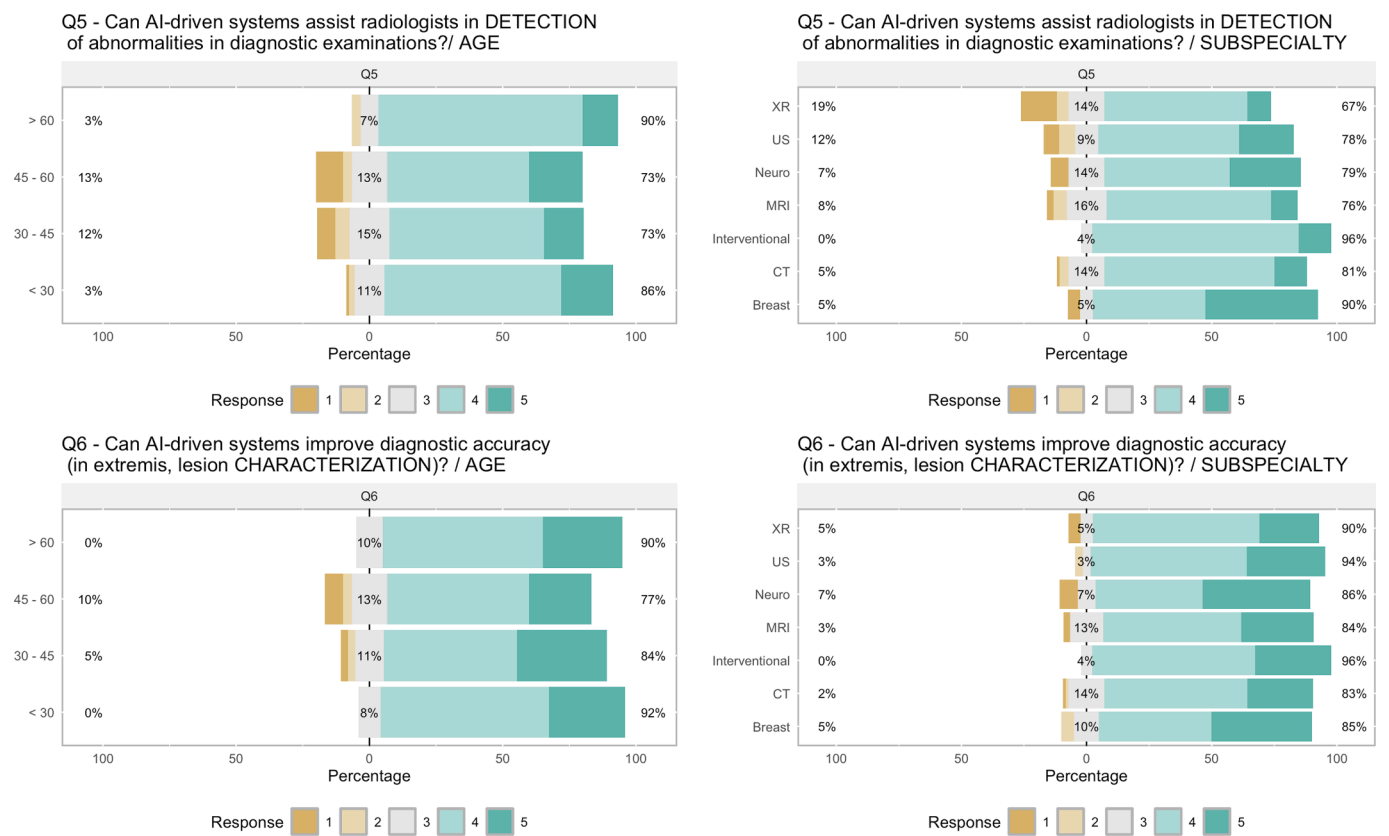


Fig. 8. Responses to question Q5 (first row) and Q6 (bottom row) stratified by age and subspecialty.

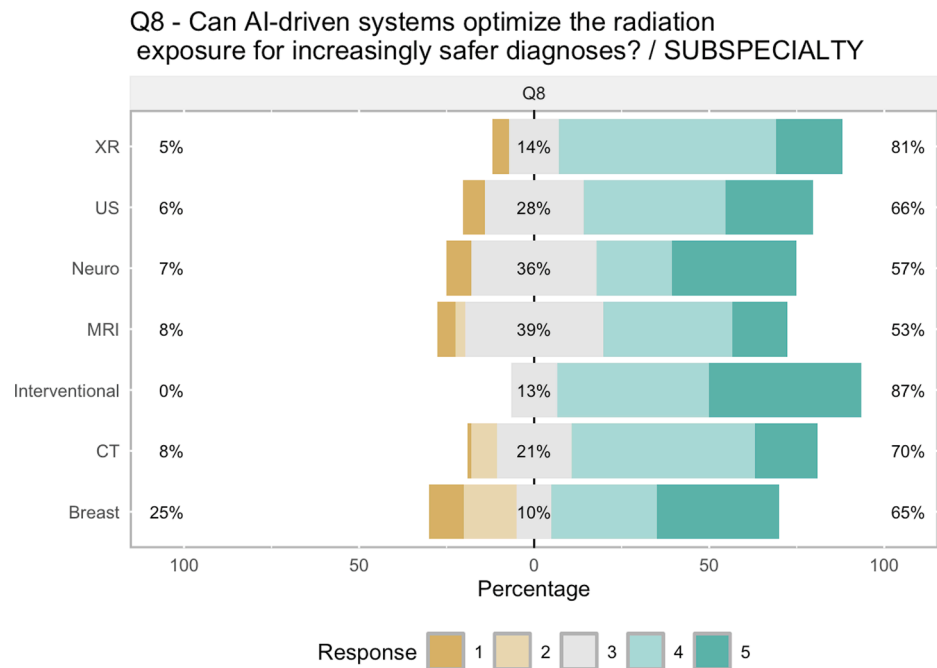


Fig. 9. Responses to question Q8 stratified by subspecialty.

these tools make a decisive contribution. These data highlight two key points: firstly, there is a general openness towards innovation within the field. Secondly, there is skepticism regarding the actual contribution of these tools in practice. This fact may derive from several factors, such as limitations in fully harnessing AI capabilities and potential overestimations of its practical impact. However, when stratifying by age subgroups, it becomes evident that senior radiologists tend to perceive this technology as more impactful overall. One hypothesis is that they perceive a greater gap represented by new technologies compared to “digital natives”, but it is certainly an aspect that deserves to be explored in greater depth.

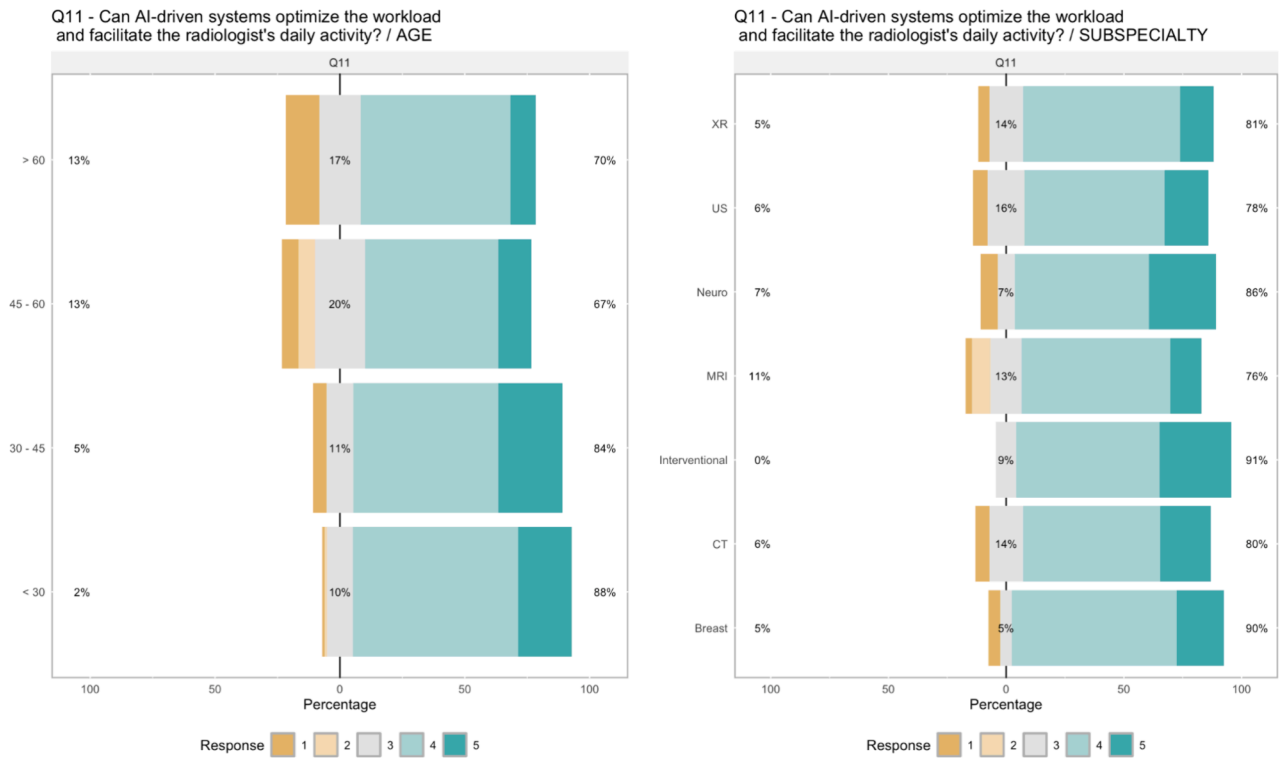


Fig. 10. Responses to question Q11 stratified by age and subspecialty.

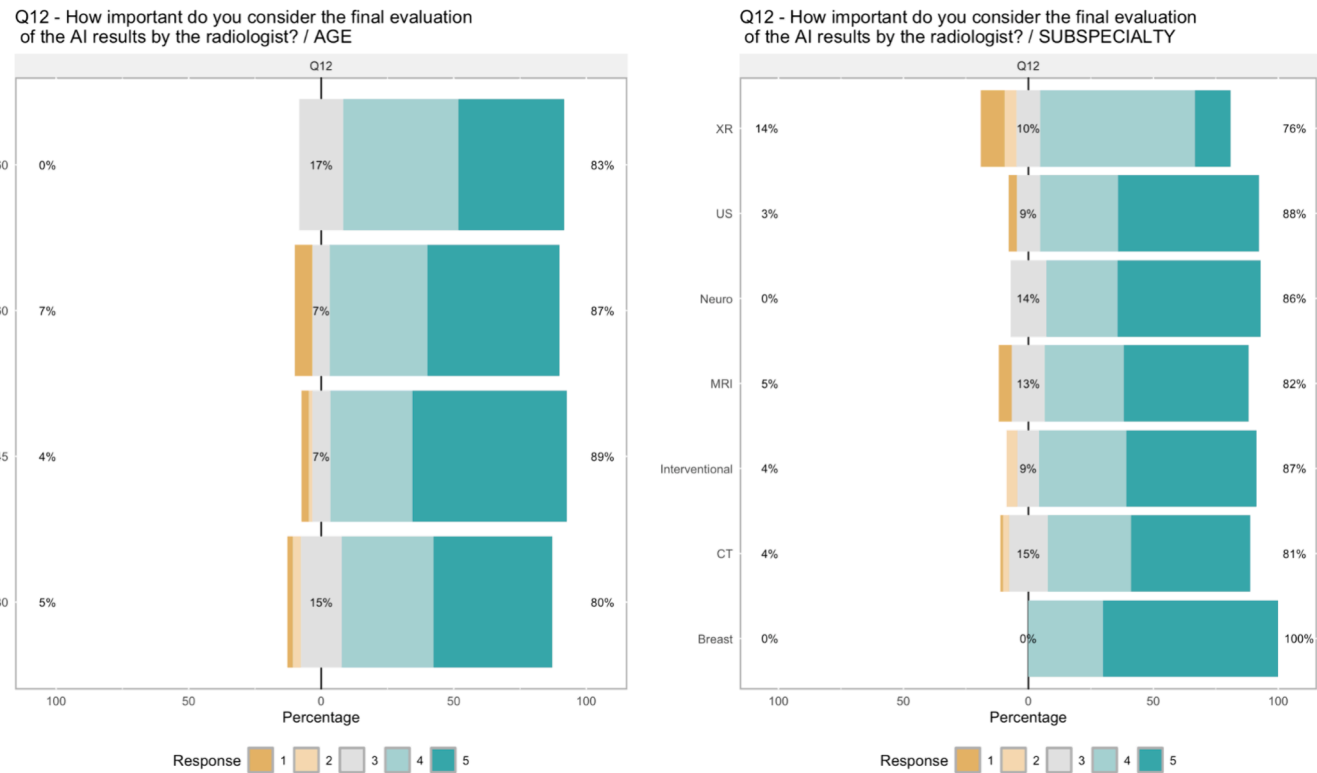


Fig. 11. Responses to question Q11 stratified by age and subspecialty.

4.2. Self-assessment of AI literacy

AI in medicine is a very complex topic and to understand all its facets and implications, multidisciplinary knowledge is required that certainly

goes beyond standard medical training. Respondents were asked to rate their preparation on the topic: 36 % considered their knowledge absent or very poor and 43 % only barely sufficient, with the most negative opinions expressed by young radiologists. Stratyfing by age group

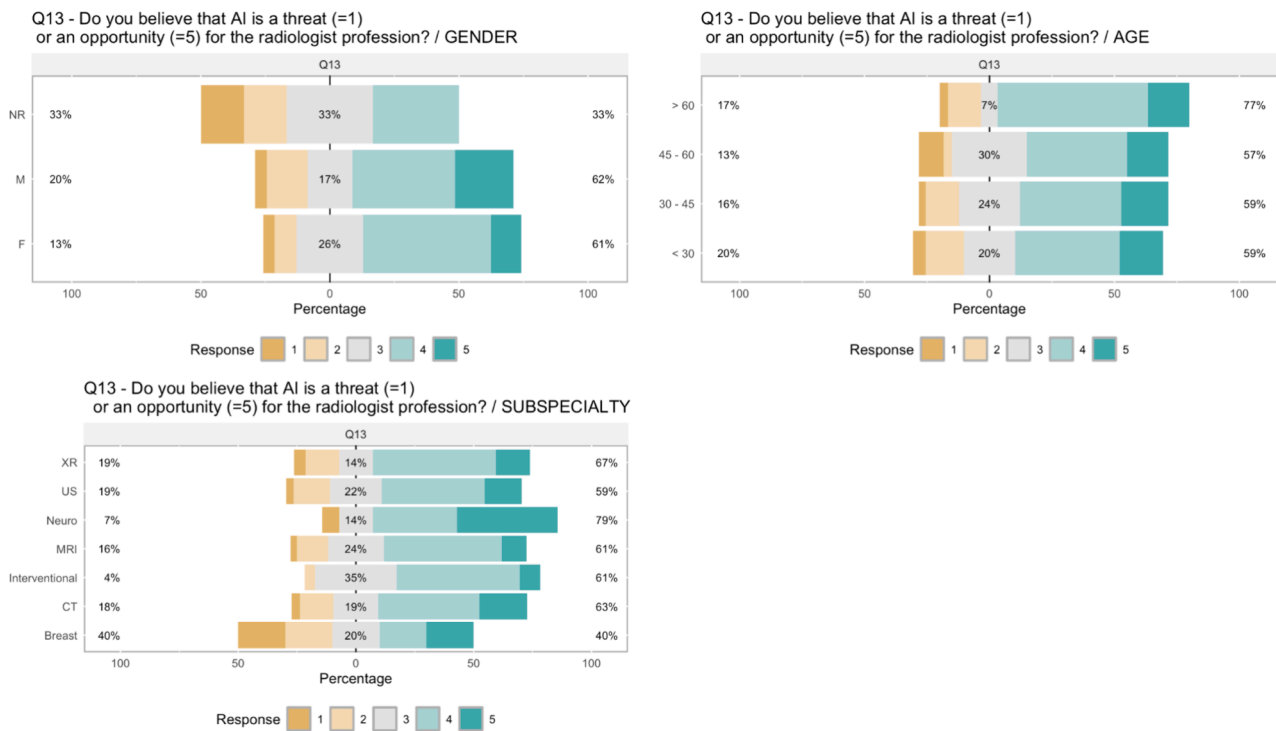


Fig. 12. Responses to question Q13 stratified by gender, age and subspecialty.

showed a trend of increasing proportion of positive responses, with subjects aged > 60 years judging their preparation to be more adequate on average compared to younger ones. Although seemingly counterintuitive, our results are in line with those produced by another survey conducted by Ooi et al. [29].

As a first hypothesis, the more negative judgment expressed by trainees can likely be attributed to their heightened awareness of the topic's complexity coupled with their relatively limited clinical experience. This combination may lead to self-underestimation when compared to senior radiologists. In contrast, senior radiologists tend to express more positive judgments, possibly due to their extensive clinical experience, which boosts their self-confidence. This confidence might lead to an overestimation of their ability to effectively integrate AI technologies. Furthermore, senior radiologists' greater openness to AI may be rooted in a pragmatic understanding of its benefits, such as reducing repetitive tasks and enhancing overall efficiency—benefits that trainees, still in the process of developing their diagnostic skills, might not fully appreciate yet. Finally, we observed no significant differences in responses by employment type or subspecialty, as previously observed in another survey assessing perspectives on AI in an Australian sample [30].

4.3. AI for enhancing diagnosis

Our data suggests a generally positive perception among radiologists of the potential role of artificial intelligence in aiding diagnostic processes. A significant percentage of respondents believe that artificial intelligence systems can effectively improve the detection (Q5) and characterization (Q6) of anomalies in radiological examinations, improving overall diagnostic accuracy and patient outcomes. These results agree with those of the annual survey conducted by the ACR Data Science Institute, which found that most respondents use artificial intelligence to improve their diagnostic performance, particularly in specific high-risk tasks such as the detection of intracranial hemorrhages, pulmonary emboli, and mammographic anomalies [31]. On the other hand, only 50 % of respondents believe that systems based on artificial intelligence can improve the radiologist's decision-making processes

regarding the diagnostic path and in particular the choice of subsequent investigations (Q7). The results are, on the other hand, in line with other studies that highlight how, although radiologists agree on the usefulness of alerting in case of anomalous findings [32], the final interpretation of the pathological findings and the choice of subsequent investigations remain the specific task of the radiologist. This may be due to limited confidence in the ability of AI to consider all the variables that govern the formulation of a diagnosis and a diagnostic process, including detailed knowledge of the patient's clinical history and preferences, relational and motivational latent variables, as well as local organizational and procedural aspects. We will return to this topic in the following paragraphs.

4.4. Artificial intelligence for personalized medicine

Most respondents (68 %) express a positive opinion on the potential role of AI in reducing radiation exposure (Q8) to meet the “as low as reasonably achievable” (ALARA) principle. The highest percentage of positive responses is recorded among interventional radiologists (87 %), the category most sensitive to the problem of dose reduction. This could be linked to the fact that those who deal with traditional radiology are generally at the beginning of the diagnostic path, and frequently involved in the problem of directing patients to the most appropriate subsequent analyses.

Contrasting opinions are obtained regarding questions Q9 and Q10, respectively on the role of AI in protocol optimization and personalization of the radiological investigation. In both questions, a neutral response pattern prevails (39 %), perhaps suggesting that, despite a generally favorable attitude towards AI-driven innovations, a full awareness of the potential of these tools in terms of personalized medicine has not yet developed [33,34]. On the other hand, many applications still seem to be the prerogative of the field of scientific research and far from widespread clinical integration.

4.5. Artificial intelligence and the profession of radiologist

The final part of our questionnaire investigated some aspects of the

challenging relationship between AI and radiologists. Regardless of the type of activity, most respondents believe that AI can improve daily workflow and reduce workload (Q11). This is an important issue in a context dominated by resource scarcity and the growing risk of burnout [35]. In general, our results are in line with those of a previous survey published by Shelmardine et al. [32], according to which respondents believed AI would positively affect their work by increasing workflow efficiencies in reporting and triage, improving diagnostic accuracy, reducing time in post-processing images and imaging radiation doses.

Despite the general positive attitude towards AI that emerged in our study, some issues of concern remain. Indeed, 84 % underlined the importance of the final radiologist's supervision of AI results to achieve optimal diagnostic accuracy. The importance of the radiologist in the final supervision of AI results has already been stated [36], and the irreplaceable role of the radiologist as an expert in the evaluation and interpretation of images is also emphasized in this investigation. It is likely, however, that this judgment includes various nuances, including not only diagnostic issues but also ethical and medico-legal implications [37]. It could be considered in the future to compare the opinions of radiologists working in different clinical settings, for example in breast cancer imaging, comparing Artificial intelligence-supported screen reading versus standard double reading [38].

Understanding whether artificial intelligence represents a threat or an opportunity for radiologists is a fundamental aspect of the current debate on technological advances in the healthcare sector. Sixty-one percent of our respondents favor the idea that AI can represent an opportunity for radiologists, they believe that AI would not diminish their professionalism. Our results are in line with a study involving pediatric radiologists, in which respondents were convinced that AI will not devalue their profession, nor lead to a reduction in the number of radiology specialists [32], and with observations from another survey conducted among European Radiologists, in which 60 % of cases do not declare any fear when asked if they were worried about a possible replacement of their work with AI [24]. In contrast, only 4 % of respondents assigned scores of 1 or 2, indicative of perceiving artificial intelligence as a threat. Importantly, this sentiment appears to be consistent across various demographic subgroups, suggesting widespread acceptance of the potential benefits of AI within the radiology community. Breast radiologists expressed relatively greater concern about AI integration, with 40 % expressing concern that AI could pose a significant threat to the professionalism and role of the radiologist. To judge whether this concern is more or less justified, we cannot ignore the fact that breast radiology is one of the areas where AI-based technology has seen the greatest implementation in clinical practice, with several AI-driven tools that have been authorized and marketed, although with regulatory differences between Europe and the United States [39].

Looking ahead, most responders agreed that the choice of specialty in the future will be increasingly influenced by AI adoption. This anticipation underscores the transformative potential of AI in shaping the future trajectory of radiological practice and underscores the need for radiologists to adapt to these technological advancements proactively.

Overall, our findings reinforce the notion that, despite apprehensions in certain quarters, the prevailing sentiment among radiologists is one of optimism regarding the transformative potential of AI within their field.

4.6. Challenges and future perspectives

As argued by Lekadir et al. [7], in the future, artificial intelligence is unlikely to replace doctors but rather to enhance their capabilities, improving diagnostic performance while freeing up resources for neglected areas such as the doctor-patient relationship. However, it's important to recognize that radiologists' concerns on specific issues are valid. This is evidenced by the apprehension among breast specialists who have experienced AI's disruptive impact, particularly fearing professional devaluation. Therefore, any further discussion must

acknowledge the awareness that we are facing an epochal paradigm shift.

Our study underscores a progressively favorable attitude towards AI integration in radiology, indicating an increasing understanding of its potential benefits. However, it also highlights the imperative need to confront challenges associated with healthcare professionals' education and readiness to effectively utilize AI [40]. Some critical points have emerged to guide future research and stimulate more in-depth reflection on these issues.

Firstly, it's crucial to contextualize our findings within a global perspective, particularly regarding the varying adoption rates observed across different geographic regions. While our results align with studies conducted in European settings, they underscore the international variability in AI adoption, with countries like China significantly more advanced in AI implementation [43].

Moreover, our findings emphasize the critical need to prioritize educational initiatives and continuous professional development across all career stages, not just among trainees. The existing training framework for radiologists may inadequately prepare them in AI fundamentals, necessitating tailored educational programs [41,42]. These initiatives should impart a comprehensive understanding of AI, covering its potential benefits and associated risks, thereby equipping radiologists to effectively navigate and harness these technologies.

Additionally, there's a compelling case for reevaluating and fostering collaboration and cross-fertilization among professionals with diverse backgrounds to fully capitalize on the opportunities presented by this paradigm shift [40]. Embracing interdisciplinary collaboration can facilitate the integration of AI into clinical practice and research activities, fostering innovation and enhancing patient care outcomes.

Finally, it's imperative to address regulatory and ethical considerations inherent in the integration of AI in radiology. As highlighted in critical literature, navigating regulatory frameworks and ethical dilemmas surrounding AI implementation is paramount to safeguarding patient safety and upholding ethical standards [28,37]. Consequently, future research endeavors should prioritize exploring and addressing these challenges to ensure responsible and equitable AI deployment in radiological practice.

4.7. Limitations

This study has several limitations. Firstly, the limited response rate may introduce selection bias, as respondents more interested in the topic were likely to have a more positive attitude towards AI. Additionally, the fact that the radiologists and residents surveyed practice in a specific region may further introduce selection bias. Due to logistical constraints, it was not feasible to interview a national or international cohort of radiologists. Nevertheless, despite its size, the opinion of our cohort of radiologists is still valuable, as it aligns with similar cases reported in the literature [40,41]. Furthermore, the assessment of knowledge levels among the radiologists who responded to this survey was entirely subjective, although consistent with findings in other studies [23].

The survey is not exhaustive. Numerous other pertinent questions could be addressed to explore the phenomenon of AI more comprehensively. For instance, examining the legal aspects of medical liability related to the use of AI tools could provide valuable insights. Although we attempted to reduce ambiguity, the questions were quite general and some could be interpreted with more nuances. More specific questions would have taken longer to complete, dramatically reducing respondent compliance. We chose to start with a short general survey to explore participants' knowledge and attitudes. Future research will involve more detailed investigations and different methodologies to deepen our understanding. To delve deeper into some topics, we plan to organize dedicated focus groups.

5. Conclusion

Overall, our findings highlight the growing acceptance and reliance on AI technologies in radiology. However, while the integration of AI tools into clinical practice is constantly growing, opinions regarding their impact remain divided.

The perceived adequacy of AI knowledge varies across different generations. Younger radiologists, who are still developing their expertise, tend to view AI as a tool that can enhance their diagnostic capabilities. In contrast, older radiologists, with established careers, often see AI as a means to maintain work quality with reduced personal effort. Intermediate age groups, in the midst of their careers, perceive AI as both an opportunity and a challenge, resulting in diverse opinions. Interestingly, despite their enthusiasm for AI, younger radiologists rated their knowledge as less adequate than more experienced professionals. Further investigation is needed to better understand the factors contributing to this generational divide.

Most radiologists believe that AI can significantly enhance diagnostic accuracy, although they are skeptical about the role of AI in personalized medicine in general, indicating this as an area that requires additional exploration and discussion. Interventional radiologists seemed more inclined to appreciate AI's benefits to their practice, even if this optimism might be due to the operational nature of their work, which makes complete replacement by AI technologies challenging. Overall, AI tools are predominantly perceived as an opportunity rather than a threat in radiology. However, a partial exception is noted among breast radiologists, probably because this subgroup is most acutely experiencing the impact of emerging AI-based technologies. Their concerns may be well-founded and deserve further consideration. Finally, most respondents agreed that AI will play a pivotal role in influencing the decisions of young doctors aspiring to enter the field of radiology.

The evolving landscape of AI in radiology presents both significant challenges and promising opportunities. AI adoption initiatives in radiology should consider these differences, providing tailored support and training for each age group to address their concerns and maximize perceived benefits.

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CRediT authorship contribution statement

Maurizio Cè: Writing – review & editing, Writing – original draft, Supervision, Investigation, Formal analysis, Data curation, Conceptualization. **Simona Ibba:** Investigation, Formal analysis, Data curation, Conceptualization. **Michaela Cellina:** Writing – review & editing, Writing – original draft, Data curation, Conceptualization. **Chiara Tancredi:** Writing – original draft, Investigation, Formal analysis, Data curation, Conceptualization. **Arianna Fantesini:** Investigation, Formal analysis, Data curation, Conceptualization. **Deborah Fazzini:** Visualization, Validation, Supervision, Project administration. **Alice Fortunati:** Validation, Supervision, Project administration, Investigation. **Chiara Perazzo:** Writing – original draft, Investigation, Formal analysis, Data curation, Conceptualization. **Roberta Presta:** Visualization, Validation, Supervision, Methodology. **Roberto Montanari:** Visualization, Validation, Supervision. **Laura Forzenigo:** Project administration, Methodology, Investigation. **Gianpaolo Carrafiello:** Writing – review & editing, Visualization, Validation, Supervision. **Sergio Papa:** Writing – review & editing, Visualization, Validation, Supervision. **Marco Ali:** Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Resources.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Supplementary material

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.ejrad.2024.111590>.

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